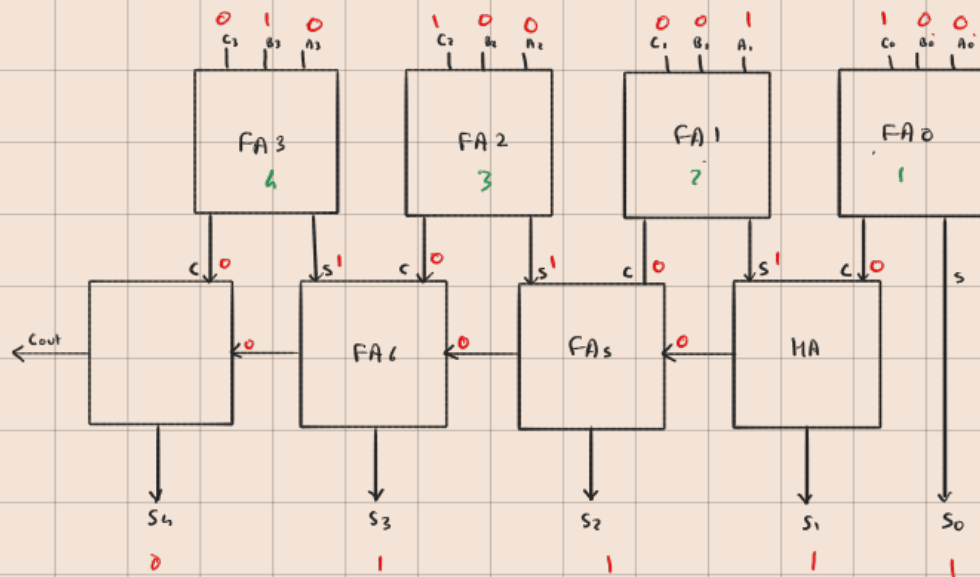


Carry Save Adder.

- Used to compute sum of 3 or more numbers together.
- Uses full Adders.
- Instead of propagating carries it stores them
- Calculates the sum and carry at the end stage.

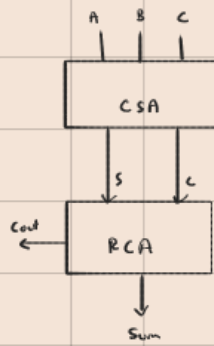
4 bit CSA



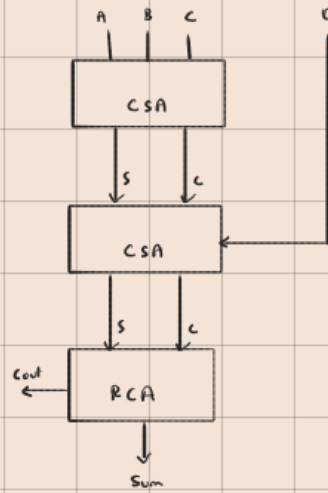
$$\begin{array}{r}
 \begin{array}{cccc}
 3 & 2 & 1 & 0 \\
 0 & 0 & 1 & 0 & A \\
 & & & & \\
 1 & 0 & 0 & 0 & B \\
 & & & & \\
 + & 0 & 1 & 0 & 1 & C \\
 \hline
 & 1 & 1 & 1 & 1
 \end{array}
 \end{array}$$

CSA for n numbers

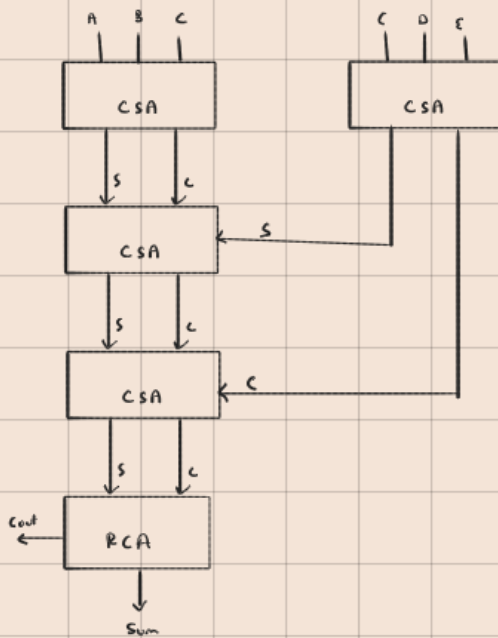
$n = 3$



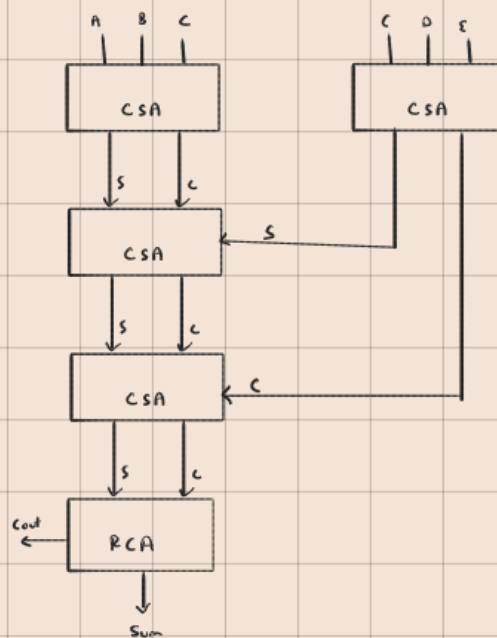
$n = 4$



$n = 6$



$n = 7$



- This Design eliminates delay because there is no propagation
- for each bit the outputs are produced parallel to each other.
- hence it has the same delay as using a single full adder.